Milestone 2

1. Project Goal

This project aims to develop an interactive data visualization system based on the QS World University Rankings dataset. Rather than presenting a simple linear ranking of universities, the objective is to explore the multidimensional structure underlying global university rankings.

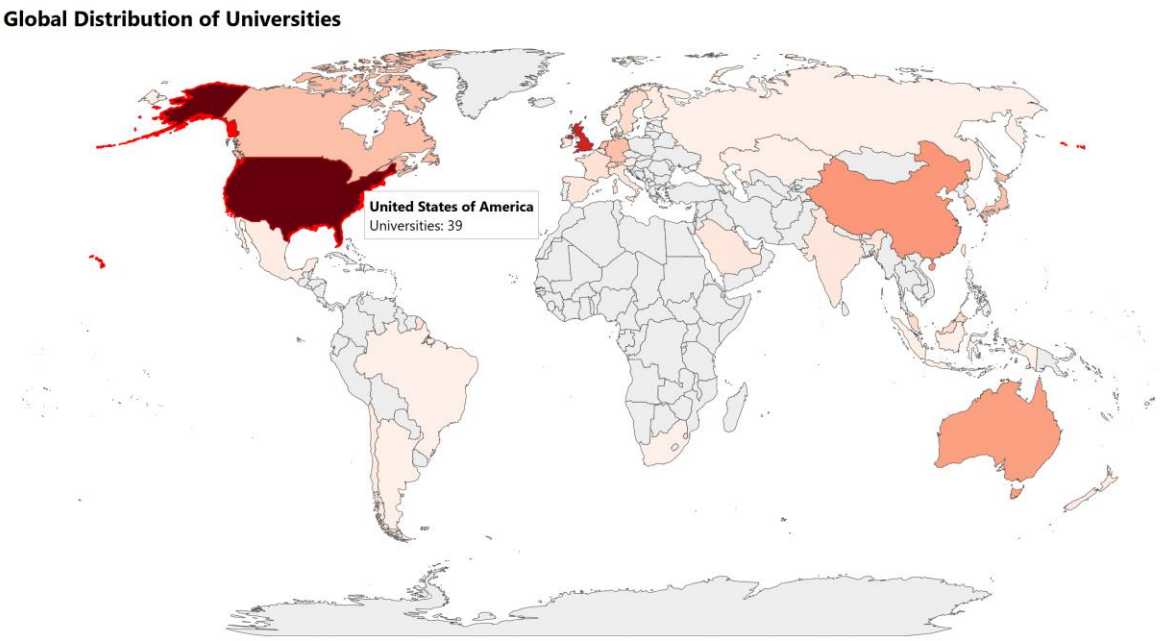
The main research questions guiding this project are the following:

- How are universities distributed across different countries and regions?
- Do universities from different regions exhibit distinct strengths across ranking indicators?
- To what extent is the overall ranking driven by a small subset of indicators?
- What are the key differences between top-ranked universities?

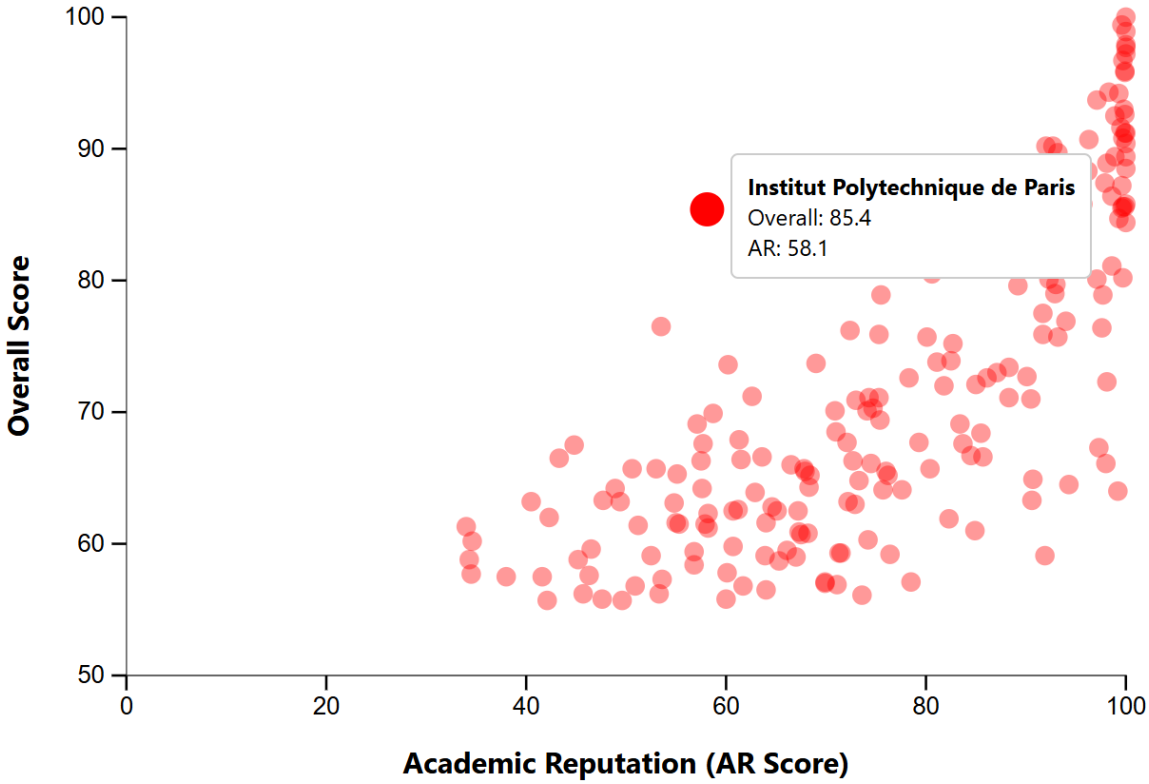
The final product will be an interactive web application that allows users to explore the dataset dynamically. Users will be able to filter universities by country or region, compare multiple institutions across several indicators, and analyze relationships between ranking metrics. Optionally, the system may also include temporal analysis to observe how rankings evolve over time.

2. Visualization Design

The first component is a **global world map**, which provides an overview of the geographical distribution of universities. This visualization will use a choropleth design, where countries or regions are colored according to the number of universities included in the QS World University Rankings. Specifically, countries with a higher number of ranked universities will be displayed with darker shades, while those with fewer universities will appear lighter. Users will be able to interact with the map by selecting a country, which will serve as a filter for the other visualizations.

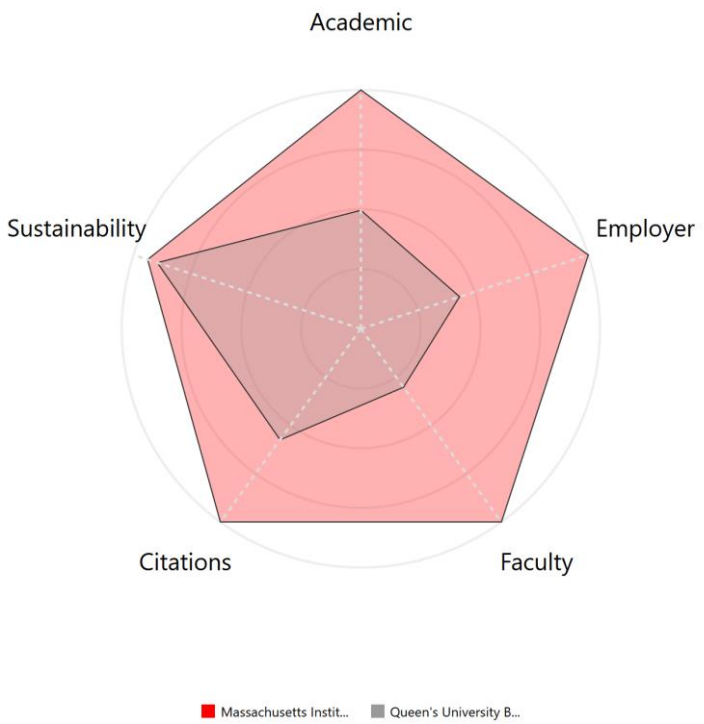


The second component is a **scatter plot**, which is used to analyze relationships between ranking indicators and overall performance. For example, one axis may represent **Academic Reputation**, while the other represents **Overall Score**. Each point corresponds to a university.

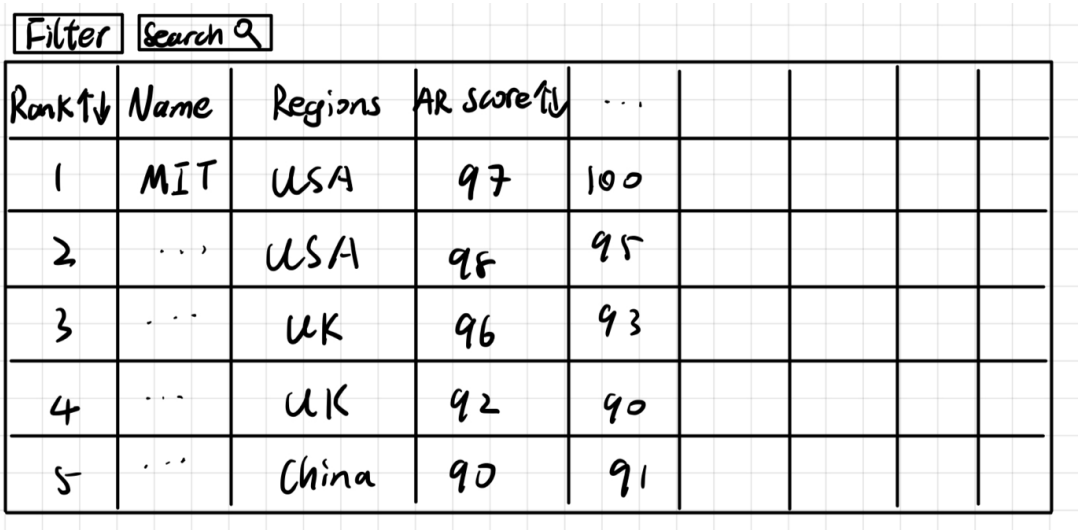


The third component is a **radar chart**, which provides a detailed view of individual universities. It displays multiple indicators such as academic reputation, employer reputation,

faculty–student ratio, citations, and internationalization. This allows users to compare the strengths and weaknesses of institutions across different dimensions.



The fourth component is a **ranking table** combined with filtering and sorting functionalities. This table allows users to search for specific universities, sort them by different indicators, and access precise numerical values. It complements the visualizations by providing exact data.



Filter Search

Rank↑↓	Name	Regions	AR score↑↓	...				
1	MIT	USA	97	100				
2	...	USA	98	95				
3	...	UK	96	93				
4	...	UK	92	90				
5	...	China	90	91				

3. Tools and Technologies

The visualizations in this project will be implemented primarily using D3.js.

The **global map** relies on D3's geographic features. The relevant lectures include: D3.js, Interactions, Marks and Channels, and Maps.

The **scatter plot** and **radar chart** are also implemented using D3.js. The relevant lectures include: Interactions, Designing Visualization, and Tabular Data.

The **ranking table** is implemented using HTML, CSS, and JavaScript. The relevant lectures include: JavaScript and Data Processing.

4. Project Breakdown

We divide the project into several independent components.

minimal viable product:

- Data cleaning and preprocessing
- Basic webpage structure (index page, header, footer)
- Main visualizations (map, scatter plot, radar chart, ranking table)
- Basic interactions (country filtering, hover information)

Optional features:

- Ranking evolution over time
- More advanced interactions and animations
- Improved website design and layout

5. Prototype

The current prototype focuses on the global map visualization. A world map is displayed using D3.js, where users can interact by clicking on different countries. This interaction allows for the selection of a specific country, which will later be used to filter and update other visualizations.

Although other components such as the scatter plot, radar chart, and ranking table have not yet been fully implemented, their structure has already been planned and partially integrated into the webpage layout.

Website link: https://com-480-data-visualization.github.io/CLZ/